Geometric Methods of mathematical Physics

Exercises.

Chapter 2

Exercise 2.1

$$\hat{r} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y} \quad , \quad \hat{\theta} = -\sin\theta\,\hat{x} + \cos\theta\,\hat{y}$$

with $\hat{x} = \partial/\partial x$, $\hat{y} = \partial/\partial y$

Show $\hat{r}$, $\hat{\theta}$ are non-coordinate bases $\Rightarrow$ show $[\hat{r}, \hat{\theta}] \neq 0$

$$\forall\ [\hat{r}, \hat{\theta}] = \hat{r}\hat{\theta} - \hat{\theta}\hat{r} = (c\hat{x} + s\hat{y})(-s\hat{x} + c\hat{y}) - (-s\hat{x} + c\hat{y})(c\hat{x} + s\hat{y})$$

where $c = \cos\theta$, $s = \sin\theta$

$$\Rightarrow [\hat{r}, \hat{\theta}] = -c\hat{x}s\hat{x} + c\hat{x}c\hat{y} - s\hat{y}s\hat{x} + s\hat{y}c\hat{y} - (-s\hat{x}c\hat{x} - s\hat{x}s\hat{y} + c\hat{y}c\hat{x} + c\hat{y}s\hat{y})$$

$$= [s\hat{x}, c\hat{x}] + [c\hat{x}, c\hat{y}] + [s\hat{x}, s\hat{y}] + [s\hat{y}, c\hat{y}]$$

$$c\hat{x}s\hat{x} = \hat{x}cs\hat{x} + c\hat{x}(s)\hat{x}$$

$$\Rightarrow [s\hat{x}, c\hat{x}] = sc\hat{x}\hat{x} + s\hat{x}(c)\hat{x} - cs\hat{x}\hat{x} - c\hat{x}(s)\hat{x}$$

$$= s\hat{x}(c)\hat{x} - c\hat{x}(s)\hat{x} \qquad \text{// use} \rightarrow \text{chain rule}$$

$$= s\frac{\partial\theta}{\partial x}\frac{\partial c}{\partial\theta}\hat{x} - c\frac{\partial\theta}{\partial x}\frac{\partial s}{\partial\theta}\hat{x} \qquad \text{// } \frac{\partial s}{\partial\theta} = c \ , \ \frac{\partial c}{\partial\theta} = -s$$

$$= -s^2\frac{\partial\theta}{\partial x}\hat{x} - c^2\frac{\partial\theta}{\partial x}\hat{x} = -\frac{\partial\theta}{\partial x}\hat{x} \qquad \text{// } s^2 + c^2 = 1$$

$$\forall\ [s\hat{x}, c\hat{x}] = -\underline{\frac{\partial\theta}{\partial x}\hat{x}}$$

Similarly $[c\hat{x}, c\hat{y}] = c\hat{x}(c)\hat{y} - c\hat{y}(c)\hat{x} = c\frac{\partial\theta}{\partial x}\frac{\partial c}{\partial\theta}\hat{y} - c\frac{\partial\theta}{\partial y}\frac{\partial c}{\partial\theta}\hat{x}$

$$= \underline{cs\left(\frac{\partial\theta}{\partial y}\hat{x} - \frac{\partial\theta}{\partial x}\hat{y}\right)}$$

$$[s\hat{x}, s\hat{y}] = s\hat{x}(s)\hat{y} - s\hat{y}(s)\hat{x} = s\left(\frac{\partial\theta}{\partial x}\frac{\partial s}{\partial\theta}\hat{y} - \frac{\partial\theta}{\partial y}\frac{\partial s}{\partial\theta}\hat{x}\right)$$

$$= \underline{-cs\left(\frac{\partial\theta}{\partial y}\hat{x} - \frac{\partial\theta}{\partial x}\hat{y}\right)}$$

$$[s\hat{y}, c\hat{y}] = s\hat{y}(c)\hat{y} - c\hat{y}(s)\hat{y} = s\frac{\partial\theta}{\partial y}\frac{\partial c}{\partial\theta}\hat{y} - c\frac{\partial\theta}{\partial y}\frac{\partial s}{\partial\theta}\hat{y}$$

$$= (-s^2 - c^2)\frac{\partial\theta}{\partial y}\hat{y} = -\underline{\frac{\partial\theta}{\partial y}\hat{y}}$$

($\frac{\partial c}{\partial\theta} = -s$, $\frac{\partial s}{\partial\theta} = c$)

Collecting the results we get

$$[\hat{r}, \hat{\theta}] = -\frac{\partial \theta}{\partial x}\hat{x} - \frac{\partial \theta}{\partial y}\hat{y} \neq 0 \quad \text{in general}$$

$\Rightarrow$ Since $[\hat{r}, \hat{\theta}]$ doesn't vanish in general it is not a coordinate basis.

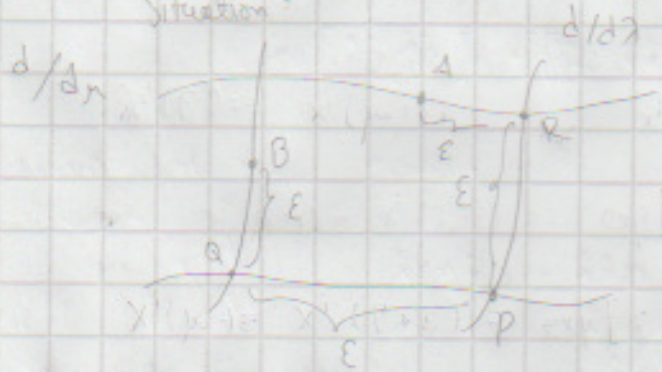
Exercise 2.2

a) Use (2.1) to prove (2.12)

(2.6): $x^i(\lambda_0 + \varepsilon) = \exp\left[\varepsilon \frac{d}{d\lambda}\right] x^i|_{\lambda_0}$

(2.12): $x^i(B) - x^i(A) = \varepsilon^2 [\vec{V}, \vec{W}] + O(\varepsilon^3)$

Situation:



$\Rightarrow$ $X(B) = X(Q+\varepsilon)$ along $d/d\lambda$

$X(Q) = X(P+\varepsilon)$ along $d/d\mu$

$X(A) = X(R+\varepsilon)$ along $d/d\mu$

$X(R) = X(P+\varepsilon)$ along $d/d\lambda$

$\Rightarrow$ $x^i(B) - x^i(A) = x^i(Q+\varepsilon) - x^i(R+\varepsilon)$ || use 2.6

$= \exp\left[\varepsilon \frac{d}{d\lambda}\right] x^i|_Q - \exp\left[\varepsilon \frac{d}{d\mu}\right] x^i|_R$ || $x^i_Q = x(Q) = X(P+\varepsilon)$

$= \exp\left(\varepsilon \frac{d}{d\lambda}\right) x^i(P+\varepsilon) - \exp\left[\varepsilon \frac{d}{d\mu}\right] x^i(P+\varepsilon)$ || Note that the two $x^i(P+\varepsilon)$ are different

$= \exp\left(\varepsilon \frac{d}{d\lambda}\right) \exp\left(\varepsilon \frac{d}{d\mu}\right) x^i|_P - \exp\left[\varepsilon \frac{d}{d\mu}\right] \exp\left(\varepsilon \frac{d}{d\lambda}\right) x^i|_P$

$= \left[\exp\left(\varepsilon\, d/d\lambda\right), \exp\left(\varepsilon\, d/d\mu\right)\right] x^i|_P$

$= \varepsilon^2 \left[\frac{d}{d\lambda}, \frac{d}{d\mu}\right] x^i|_P + O(\varepsilon^3)$ || $\vec{V} = \frac{d}{d\lambda} x^i|_P$, $\vec{W} = \frac{d}{d\mu} x^i|_P$

$= \varepsilon^2 [\vec{V}, \vec{W}] + O(\varepsilon^3)$

(b) Prove that $\exp\left[a\frac{d}{d\lambda} + b\frac{d}{d\mu}\right] = \exp[ad/d\lambda]\exp[bd/d\mu]$ (1)

for all a & b iff $[d/d\lambda, d/d\mu] = 0$

Summation is commutative i.e; $a\frac{d}{d\lambda} + b\frac{d}{d\mu} = b\frac{d}{d\mu} + a\frac{d}{d\lambda}$

(but multiplications of vectors isn't necessarily.

$\Rightarrow \left[\exp\left[a\frac{d}{d\lambda}\right], \exp[bd/d\mu]\right] = \exp\left(a\frac{d}{d\lambda}\right)\exp\left(b\frac{d}{d\mu}\right) - \exp\left(b\frac{d}{d\mu}\right)\exp\left(a\frac{d}{d\lambda}\right)$

Assuming (1) holds:

$\left[\exp(a\,d/d\lambda), \exp(b\,d/d\mu)\right] = \exp\left(a\frac{d}{d\lambda} + b\frac{d}{d\mu}\right) - \exp\left(b\frac{d}{d\mu} + a\frac{d}{d\lambda}\right) = 0$

$\Rightarrow$ So (1) working implies $[d/d\lambda, d/d\mu] = 0$, but at the same time if (1) works but we assume $[d/d\lambda, d/d\mu] \neq 0$ then we get an contradiction

$\Rightarrow \exp\left[a\frac{d}{d\lambda} + b\frac{d}{d\mu}\right] = \exp\left[a\frac{d}{d\lambda}\right]\exp\left[b\frac{d}{d\mu}\right]$ iff $\left[\frac{d}{d\lambda}, \frac{d}{d\mu}\right] = 0$

Another way to see this is to expand the exponentials of (1) as Taylor series & compare both sides.

Exercise 2.3 Prove that any C^3 vector fields $\bar{X}, \bar{Y}, \bar{Z}$ satisfy the Jacobi identity:

$$\left[[\bar{X}, \bar{Y}], \bar{Z}\right] + \left[[\bar{Y}, \bar{Z}], \bar{X}\right] + \left[[\bar{Z}, \bar{X}], \bar{Y}\right] = 0$$

This is quite trivial

$$0 = [\bar{X},\bar{Y}]\bar{Z} - \bar{Z}[\bar{X},\bar{Y}] + [\bar{Y},\bar{Z}]\bar{X} - \bar{X}[\bar{Y},\bar{Z}] + [\bar{Z},\bar{X}]\bar{Y} - \bar{Y}[\bar{Z},\bar{X}]$$

$$= [\bar{X}\bar{Y} - \bar{Y}\bar{X}, \bar{Z}] + [\bar{Y}\bar{Z} - \bar{Z}\bar{Y}, \bar{X}] + [\bar{Z}\bar{X} - \bar{X}\bar{Z}, \bar{Y}]$$

$$= \bar{X}\bar{Y}\bar{Z} - \bar{Y}\bar{X}\bar{Z} - \bar{Z}\bar{X}\bar{Y} + \bar{Z}\bar{Y}\bar{X} + \bar{Y}\bar{Z}\bar{X} - \bar{Z}\bar{Y}\bar{X} - \bar{X}\bar{Y}\bar{Z} + \bar{X}\bar{Z}\bar{Y} + \bar{Z}\bar{X}\bar{Y} - \bar{X}\bar{Z}\bar{Y} - \bar{Y}\bar{Z}\bar{X} + \bar{Y}\bar{X}\bar{Z}$$

// We have terms that cancel with same symbol above or below

$= 0$, as all terms cancel □. // Jacobi identity holds

Exercise 2.4

A matrix is $\binom{1}{1}$-tensor so a linear transformation T is a map

T ($\binom{1}{1}$-tensor) $\to$ $\binom{1}{1}$-tensor, i.e it spits out one one form and one vector when given as arguments one one form and one vector. As $\binom{1}{1}$-tensor is a map that takes one one form & one vector & gives Real number we say that when T is given 2 one forms & 2 vectors as arguments it gives real number as a value. It is a bilinear function which takes as arguments 2 one-forms an 2 vectors and whose value is a real number. This is a definition of $\binom{2}{2}$-tensor $\Rightarrow$ T is $\binom{2}{2}$ tensor

Exercise 2.5

a) A general $\binom{2}{0}$ tensor has n^2 independent components, meanwhile a vector has n-components. $\Rightarrow$ Outer product of two vectors has $2n$ components of which only $2n-1$ are independent because of rescaling $\frac{1}{a}V \to aW$ for $T = V \otimes W$, V, W are vectors

So we have n^2 for general $\binom{2}{0}$ tensor & $2n-1$ for outer product of two vectors and of such these vector outer product don't describe a general $\binom{2}{0}$ tensor.

b) $\vec{V}, \vec{w} \to \vec{V} \otimes \vec{w}$ components are

$$\vec{V} \otimes \vec{w} = \vec{V}^i \frac{\partial}{\partial x^i} \otimes w^j \, dx_j \quad //$$

$$= V^i w^j \left(\frac{\partial}{\partial x^i} \otimes dx_j\right)$$

$V^i w^j$: Component; $\frac{\partial}{\partial x^i} \otimes dx_j$: basis vector outer product

Exercise 2.6

Proof: Set of all $\binom{2}{0}$ tensors of P is a vector space under addition defined by analogy with

$$(\tilde{\omega} + \tilde{\sigma})(\vec{v}) = \tilde{\omega}(\vec{v}) + \tilde{\sigma}(\vec{v})$$

For analogous equations for T, R $\binom{2}{0}$ tensor would be:

$$(T + E)(\tilde{\omega}, \tilde{\sigma}) = T(\tilde{\omega}, \tilde{\sigma}) + E(\tilde{\omega}, \tilde{\sigma})$$

(Since $T \in R^n \times R^n$, $E \in R^n \times R^n$ then $T + E \in R^n \times R^n$ (2) And the multiplication by scalar is defined or like in vector are:
$a(T+E) = aT + aE$, $(a+b)T = aT + bT$, $ab T = a(bT)$, $1 \cdot T = T$
$\Rightarrow$ A set of all $\binom{2}{0}$ tensors of P is a vector space.)

Now to calculate the basis we use similar technique as in eq. (2.24) - (2.26):

$$T(\tilde{\omega}, \tilde{\sigma}) = T(w_i \tilde{\omega}^i, \sigma_j \tilde{\omega}^j) = w_i \sigma_j T(\tilde{\omega}^i, \tilde{\omega}^j)$$

// $\tilde{\omega}^i$ is the basis of one-form $\binom{2}{0}$ tensor maps it one form to real number

$$\Rightarrow T(\tilde{\omega}, \tilde{\sigma}) = \tilde{e}_i(\tilde{\omega})\, \tilde{e}_j(\tilde{\sigma})\, T(\tilde{\omega}^i, \tilde{\omega}^j)$$

$\Rightarrow$ $T(\tilde{\omega}^i, \tilde{\omega}^j)$ are the components of T

$$\Rightarrow T(\tilde{\omega}, \tilde{\sigma}) = T^{ij}\, \tilde{e}_i(\tilde{\omega})\, \tilde{e}_j(\tilde{\sigma})$$

// $\tilde{\omega}, \tilde{\sigma}$ are arbitrary

$$\Rightarrow T = T^{ij}\, \tilde{e}_i \otimes \tilde{e}_j \quad \square$$

The basis of $\binom{2}{0}$ tensor is $\tilde{e}_i \otimes \tilde{e}_j$ i.e. the set of all $\binom{2}{0}$ tensors is a $T_p \otimes T_p$ space.

Exercise 2.7

$Q^{ijk}{}_{lm}$ has 2 contraction for each upstair indices so $3 \cdot 2 = 6$ $\binom{1}{1}$ tensors may be made. AFTER this we have tensor of form $Q^{ijk}{}_{im}$

The second contraction can be done $2 \cdot 1 = 2$ times from $\binom{1}{1}$ tensor meaning from initial $\binom{3}{2}$ tensors we can be made $6 \cdot 2 = 12$ $\binom{1}{0}$ tensors by a second contraction.

Exercise 2.8

A, B are (1) tensors i.e. vector-valued (linear) functions of vectors

We define $C(\bar{v}) \equiv B(A(\bar{v}))$

This is a (1) tensor because $A(\bar{v}) = \bar{w}$ is a vector meaning

$C(\epsilon\bar{v}) = B(\bar{w})$ and B is (1) tensor i.e. vector valued linear function of vectors

$A(\bar{v}) = \sum_i A^i_j v^j e_i = A^i_j \tilde{\omega}^j(\bar{v})\, \bar{e}_i$ // A^i_j is the component

$\Rightarrow A = A^i_j\, \bar{e}_i \otimes \tilde{\omega}^j$ Similarly $B = B^i_j\, \bar{e}_i \otimes \tilde{\omega}^j$

$\Rightarrow C(\bar{v}) = B(A(\bar{v})) = B(A^i_j\, \tilde{\omega}^j(\bar{v})\, \bar{e}_i)$

$= A^i_j\, \tilde{\omega}^j(\bar{v})\, B(\bar{e}_i) = A^i_j\, \tilde{\omega}^j(\bar{v})\, B^k_\ell\, \underbrace{\tilde{\omega}^\ell(\bar{e}_i)}_{=\delta^\ell_i}\, \bar{e}_k$

$C(\bar{v}) = A^i_j B^k_i\, \tilde{\omega}^j(\bar{v})\, \bar{e}_k$ // this is independent of $\bar{v}$

$\Rightarrow C = A^i_j B^k_i\, \tilde{\omega}^j \otimes \bar{e}_k$

$\Rightarrow C^i_j = B^i_k A^k_j$ // bit of renaming

This amounts to component of matrix product when comparing with the linear transformation defined in §1.6 : $(BA)_{ij} = \sum_k B_{ik} A_{kj}$ (1.31)

Exercise 2.9 Show that $\binom{2}{0}$ tensors components transform as two vectors i.e $T^{i'j'} = \Lambda^{i'}_{k} \Lambda^{j'}_{l} T^{kl}$

$T = T^{ij}\ \bar{e}_i \otimes \bar{e}_j$ for $\binom{2}{0}$ T

In the new basis $\bar{e}_{i'} = \Lambda^{j}_{i'}\ \bar{e}_j$ thus yet

$$T = T^{i'j'}\ \bar{e}_{i'} \otimes \bar{e}_{j'} = T^{i'j'} \left(\Lambda^{i}_{i'}\ \bar{e}_i \otimes \Lambda^{j}_{j'}\ \bar{e}_j \right)$$

$$= T^{i'j'}\ \Lambda^{i}_{i'}\ \Lambda^{j}_{j'}\ \bar{e}_i \otimes \bar{e}_j = T^{ij}\ \bar{e}_i \otimes \bar{e}_j$$

$$\Rightarrow T^{i'j'}\ \Lambda^{i}_{i'}\ \Lambda^{j}_{j'} = T^{ij}$$

$$\Rightarrow T^{i'j'} = \Lambda^{i'}_{i}\ \Lambda^{j'}_{j}\ T^{ij} = \Lambda^{i'}_{k}\ \Lambda^{j'}_{l}\ T^{kl} \quad \square$$

For $\binom{N}{N'}$ tensor we have $T = T^{ij\ldots N}_{kl\ldots N'}\ \bar{e}_i \otimes \bar{e}_j \otimes \cdots \bar{e}_N \otimes \tilde{\omega}^k \otimes \cdots \tilde{\omega}^l$

- Each basis vector transforms as $\bar{e}_{i'} = \Lambda^{j}_{i'}\ \bar{e}_j$ giving $\bar{e}_j = \Lambda^{i'}_{j}\ \bar{e}_{i'}$
and the basis one-form transforms as $\tilde{\omega}^{k'} = \Lambda^{k'}_{k}\ \tilde{\omega}^k$ giving $\tilde{\omega}^k = \Lambda^{k}_{i'}\ \tilde{\omega}^{i'}$

Thus under coord transformation we get

$$T^{i'j'k'\ldots N'}_{l'm'\ldots N'} = T^{ijk\ldots N}_{lm\ldots N'}\ \Lambda^{i'}_{i}\ \Lambda^{j'}_{j} \ldots \Lambda^{N'}_{N}\ \Lambda^{k}_{k'}\ \Lambda^{l}_{l'}\ \Lambda^{m}_{m'} \ldots \Lambda^{N'}_{N'(\ldots)}$$

So each basis vector or one form contributes 1 term of linear transformation Λ

Exercise 2.10

(a) $T^{i'j'\ldots N'}_{k'l'\ldots M'} = T^{ij\ldots N}_{kl\ldots m}\ \Lambda^{i'}_{i}\ \Lambda^{j'}_{j} \ldots \Lambda^{N'}_{N}\ \Lambda^{k}_{k'}\ \Lambda^{l}_{l'} \ldots \Lambda^{m}_{m'}$

(b) if $T^{ij\ldots N}_{kl\ldots m}$ are all zero then no linear transformation from above can make $T^{i'j'\ldots N'}_{k'l'\ldots M'}$ non zero since $0 \cdot \Lambda^{i}_{j} = 0$

- If all components are zero in one basis they are zero in all basis.

Exercise 2.11

$\{A^i_j\,,\ i,j = 1,\dots,n\}$ in n-dim vector space

We define the transformation to be $A^{i'}_{k'} = \Lambda^{i'}_{j}\,\Lambda^{k}_{k'}\,A^{j}_{k}$

Show that there are components of A on new basis $\{\bar{e}_{j'}\}$

Show that A is a tensor

Tensor: tensor of type $\binom{N}{M}$ if P is defined to be a linear function which takes as arguments, M vectors and N one-forms and whose value is a real number

A^i_j transforms same as $\binom{1}{1}$-tensor as defined in exercise 2.9 so the transformation under change of basis is not an obstacle to be tensor for A^i_j (components of)

Now we only need to show the mapping $A(\tilde{w}, \bar{v})$ is a Real number for it to be tensor.

We only know the components and that A^i_j is in n-dim vector space such that in associated basis $\{\bar{e}_i\}$ the components are A^i_j while in $\{\bar{e}_{j'}\}$ $A^{i'}_{j'} = \Lambda^{i'}_{j}\,\Lambda^{j}_{j'}\,A^{i}_{k}$. The basis have relation

$\bar{e}_{j'} = \Lambda^{i}_{j'}\,\bar{e}_i$ meaning the transformation $A^{i'}_{j'} = \Lambda^{i'}_{j}\,\Lambda^{j}_{j'}\,A^{i}_{k}$

Associates with one basis vector and one one-form $\tilde{w}^{i'} = \Lambda^{i'}_{k}\,\tilde{w}^{i}$

$\Rightarrow$ We can write A in terms of outer products of $\bar{e}_i$ and $\tilde{w}^i$

$\Rightarrow$ $A = A^i_j\ \tilde{w}^j \otimes \bar{e}_i$ This is a tensor as it is a linear function which takes as argument 1-vector & 1-one-form and gives Real number $\Rightarrow$ Both definitions are define same tensor $\square$

Exercise 2.12

Find matrices Λ which cast the following matrices into their unit diagonal form.

2) Let's use eigenvalues to find Λ. $\det(A - \lambda I) = 0$ ← eigenvalue eq

$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \Rightarrow \det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = 0$

$\Rightarrow \lambda = 1$, $\lambda = 3$ So we can find $Q^T A Q = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$

Using the spectral theorem: Every real symmetric matrix $A = A^T$ can be diagonalized using an orthogonal matrix Q such that

$$Q^T A Q = D \quad , \text{ where } Q^T Q = I \quad , \; D \text{ is diagonal}$$

with $D = \text{diag}(\lambda_1 \dots \lambda_n)$ and Q is composed from orthogonal eigenvectors

Let's find the eigenvectors of A using $Av = \lambda v$

$$\Rightarrow (A - \lambda_1 I)\, v = 0 \quad \Rightarrow \left[\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \right] \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \quad \Rightarrow \quad x = y$$

$$\Rightarrow \quad v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \text{for} \qquad \lambda = 3$$

$$(A - \lambda_2 I)\, v = 0 \qquad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \quad x = -y$$

$$\Rightarrow \quad v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \qquad \text{for} \qquad \lambda_2 = 1$$

$$\Rightarrow \quad Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\Rightarrow \quad Q^T A Q = D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \qquad \| \text{ to get identity we need to rescale } v_1$$

Let's define $H = \begin{pmatrix} 1/\sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}$

$$\Rightarrow \quad \Lambda = QH \quad \text{gets us the unit diagonal form}$$

$$\Rightarrow \quad \Lambda = \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{3} & 1 \\ 1/\sqrt{3} & -1 \end{pmatrix} \qquad \Rightarrow \quad \Lambda^T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} \\ 1 & -1 \end{pmatrix}$$

$$\Rightarrow \quad \Lambda^T A \Lambda = \frac{1}{2} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} & 1 \\ 1/\sqrt{3} & -1 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1/\sqrt{3} + 3/\sqrt{3} & 1 + 3/\sqrt{3} \\ 2/\sqrt{3} + 1 & 1 - 1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} & 1 \\ 1/\sqrt{3} & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \frac{1}{3}(3+3) & 0 \\ 0 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

So $\Lambda = \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{3} & 1 \\ 1/\sqrt{3} & -1 \end{pmatrix}$ can't $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ to it's unit diagonal form

b) $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ eigenvalues: $\begin{vmatrix} \lambda & 1 \\ 1 & \lambda \end{vmatrix} = \lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$

$\lambda_1 = 1 \Rightarrow (A - \lambda_1 I) v_1 = 0$

$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x = y \Rightarrow v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\lambda_2 = -1 \Rightarrow (A - \lambda_2 I) v_2 = 0$

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x = -y \Rightarrow v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\Rightarrow Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ for $D = \text{diag}(1, -1)$

We get Λ by rescaling v_2: $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$\Rightarrow \Lambda = QH = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

$\Rightarrow \Lambda^T A \Lambda = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ <= this is the unit diagonal of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Note: we don't actually need $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ since we cannot use it to change the sign of unit diagonal so that works with just $\Lambda = Q$

c) $A = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix} \Rightarrow$ eigenvalues $0 = \begin{vmatrix} 4-\lambda & 0 \\ 0 & -1-\lambda \end{vmatrix} \Rightarrow -(4-\lambda)(1+\lambda) = 0$

$\Rightarrow \lambda_1 = 4, \lambda_2 = -1$

$\lambda_1 = 4 \Rightarrow (A - \lambda_1 I) v_1 = \begin{pmatrix} 0 & 0 \\ 0 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow y = 0, x = \text{free}$

$\Rightarrow v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$\lambda_2 = -1 \Rightarrow (A - \lambda_2 I) v_2 = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x = 0, y = \text{free}$

$\Rightarrow v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\Rightarrow Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$ for $D = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}$

$\Rightarrow$ to get unit diagonal we rescale: $H = \begin{pmatrix} \frac{1}{\sqrt{4}} & 0 \\ 0 & 1 \end{pmatrix}$

$\Rightarrow \Lambda = QH = H = \begin{pmatrix} 1/\sqrt{4} & 0 \\ 0 & 1 \end{pmatrix}$

$\Rightarrow \Lambda^T A \Lambda = \begin{pmatrix} 1/\sqrt{4} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} +4 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{4} & 0 \\ 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$

So $\Lambda = \begin{pmatrix} 1/\sqrt{4} & 0 \\ 0 & 1 \end{pmatrix}$. For this we should just recognized that A was already diagonal to it.

Exercise 2.8

a) $g^{-1}(\tilde{v}, \tilde{w}) = \tilde{v} \cdot \tilde{w} \in \mathbb{R}$ so g^{-1} $\binom{0}{2}$ tensor

Let's look at how g^{-1}'s components transform one-forms by looking at components:

$g^{-1}(\tilde{w}, \tilde{x}) = g^{ij} w_i x_j \in \mathbb{R}$ $\Rightarrow g^{-1}$ maps 2 one forms to Real number

$\Rightarrow g^{-1}$ is bilinear map of one-forms with components g^{ij}

b) $g^{-1}(\tilde{e}^i, \tilde{e}^j) = g^{ij} = \pm \delta^{ij}$ since $\{\tilde{e}^i\}$ form orthonormal basis

$\Rightarrow$ We can write g^{-1} as $g^{-1} = g^{ij} \tilde{w}^i \otimes \tilde{w}^j$ // Dual basis: $\tilde{w}^i(\tilde{e}_j) = \delta^i_j$

$\Rightarrow g^{-1}(\tilde{e}^i, \tilde{e}^j) = g^{ek} \tilde{w}^e(\tilde{e}^i) \tilde{w}^k(\tilde{e}^j)$

$= g^{ek} \delta^e_i \delta^k_j$ // Since g^{-1} is canonical in this basis $g^{ek} = \pm \delta^{ek}$

$\Rightarrow$

$= \pm \delta^{ij}$

Similarly due to gl being canonical so is gl^{-1} with only diagonal entries being non-zero and ± 1

$\Rightarrow$ $gl^{-1} = g^{ij}\, \tilde{e}_i \otimes \tilde{e}_j$ is ~~the~~ inverse of gl

$\Rightarrow$ $gl^{-1}(\tilde{\omega}^i, \tilde{\omega}^j) = g^{kl}\, \tilde{e}_k(\tilde{\omega}^i)\, \tilde{e}_l(\tilde{\omega}^j)$

$= (\pm \delta^{kl})\, \delta^i_k\, \delta^j_l = \pm \delta^{ij}$ □

$\Rightarrow$ $\{\tilde{\omega}^i\}$ is an orthonormal basis

Exercise 2.14

Let's start from general diagonal metric g (since we can always diagonalise g)

$\Rightarrow$ $g = \mathrm{diag}(f_1(p), \ldots, f_n(p)) = \mathrm{diag}(f_1, \ldots, f_n)$ for some C^∞ function f of P

We can do the transformation at point P to find a basis where g is canonical. In this case $\Lambda = \mathrm{diag}(1/\sqrt{f_1}, \ldots, 1/\sqrt{f_n})$ gives

$$g' = \Lambda^T g \Lambda = \mathrm{diag}(\pm 1, \ldots, \pm 1)$$ where the sign of each entry depends on original g.

(i) $\Rightarrow$ $g'_{ij} = \pm \delta_{ij}$ □ (so this basis where $g \to g'$ at P is orthonormal)

(ii) condition $\dfrac{\partial g_{ij}}{\partial x_k}\Big|_p = \dfrac{\partial}{\partial x_k}\Big|_p \left\{ \dfrac{1}{\sqrt{f}}\, f\, \dfrac{1}{\sqrt{f}} \right\}$ for each f in g

ignore that we let x_k instead of x^k

$\Rightarrow$ $\dfrac{\partial g_{ij}}{\partial x_k}\Big|_p = -\frac{1}{2} f^{-3/2} \left(\dfrac{\partial f}{\partial x_k}\right) f^{-1} f^{-1/2} + f^{-1/2} \left(\dfrac{\partial f}{\partial x_k}\right) f^{-1/2} - \frac{1}{2} f^{-1/2} f \left(\dfrac{\partial f}{\partial x_k}\right) f^{-3/2}\Big|_p$

$= -f^{-1}\left(\dfrac{\partial f}{\partial x_k}\right) + f^{-1}\left(\dfrac{\partial f}{\partial x_k}\right)\Big|_p = 0$ □

For (iii)-condition we just continue with additional derivation

$$\frac{\partial^2 g_{ij}}{\partial x^k \partial x^l}\Big|_p = \frac{\partial^2}{\partial x^k x^l}\left\{ f^{-1/2} f \; f^{-1/2} \right\} = \Big|_p \left(\frac{\partial^2 f^{-1/2}}{\partial x^k \partial x^l} \right) f f^{-1/2} + f^{-1/2} \left(\frac{\partial^2 f}{\partial x^k \partial x^l} \right) f^{-1/2} + f^{-1/2} f \left(\frac{\partial^2 f^{-1/2}}{\partial x^k \partial x^l} \right) \Big|_p$$

$$\frac{\partial^2 g_{ij}}{\partial x^k \partial x^l}\Big|_p = 2 f^{1/2} \left(\frac{\partial^2 f^{-1/2}}{\partial x^k \partial x^l} \right) + f^{-1} \frac{\partial^2 f}{\partial x^k \partial x^l}\Big|_p$$

This can be non-zero ⇒ Not necessarily zero ⇒

⇒ a C^∞ metric know is locally flat

Exercise 2.15

Euclidean metric is $g = \mathrm{diag}(1, 1)$ in Cartesian coords with basis $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right)$

a) new basis $\partial/\partial r$, $\partial/\partial\theta$ → $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$

$r = \sqrt{x^2 + y^2}$, $\tan\theta = \frac{y}{x}$ ⇒ $\theta = \arctan\left(\frac{y}{x}\right)$ $\{0 \le \theta < \pi$

⇒ Vector transformation under coordinate change: $\partial/\partial r = \frac{\partial x_i}{\partial r}\frac{\partial}{\partial x_i}$

$\partial/\partial\theta = \frac{\partial x_i}{\partial\theta}\frac{\partial}{\partial x_i}$

⇒ $\Lambda^{i'}_{\ j} = \frac{\partial x^{i'}}{\partial x^j}$ where $x^{i'} = (r, \theta)$ i.e primed are new coordinates

The inverse metric of this is $\Lambda^{i}_{\ j'} = \frac{\partial x^i}{\partial x^{i'}}$ $x^{i'} = (r, \theta)$

The metric transforms as: $g_{i'j'} = \Lambda^i_{\ i'} \Lambda^j_{\ j'} g_{ij}$

⇒ $g_{rr} = \frac{\partial x^i}{\partial r}\frac{\partial x^j}{\partial r} g_{ij} = \left(\frac{\partial x}{\partial r}\right)^2 g_{11} + \left(\frac{\partial y}{\partial r}\right)^2 g_{22}$ $\Big\|$ $\frac{\partial x}{\partial r} = \left(\frac{\partial r}{\partial x}\right)^{-1} = \cos\theta$

$\frac{\partial y}{\partial r} = \left(\frac{1}{x}\frac{1}{r} g_y\right)^{-1} = \sin\theta$

$= \cos^2\theta + \sin^2\theta = 1$

$\frac{\partial x}{\partial\theta} = -r\sin\theta$

$\frac{\partial y}{\partial\theta} = r\cos\theta$

$g_{\theta r} = g_{r\theta} = \frac{\partial x^i}{\partial r}\frac{\partial x^j}{\partial\theta} g_{ij} = \frac{\partial x}{\partial r}\frac{\partial x}{\partial\theta} + \frac{\partial y}{\partial r}\frac{\partial y}{\partial\theta}$

$= -r\cos\theta\sin\theta + r\sin\theta\cos\theta$

$= 0$

$$g_{\theta\theta} = \frac{\partial x^i}{\partial \theta}\frac{\partial x^j}{\partial \theta} g_{ij} = \left(\frac{\partial x}{\partial \theta}\right)^2 g_{11} + \left(\frac{\partial y}{\partial \theta}\right)^2 g_{22}$$

$$= r^2(\sin^2\theta + \cos^2\theta) = r^2$$

$$\Rightarrow \quad g = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}$$ is the Euclidean metric in polar coords $(\partial_r, \partial_\theta)$

b) $\hat{r} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y}$ $\quad \| \ \hat{x} = \partial/\partial x, \ \partial/\partial y = \hat{y}$

$\hat{\theta} = -\sin\theta\,\hat{x} + \cos\theta\,\hat{y}$

$$\Rightarrow \quad \hat{r} \ \| \ \cos\theta\frac{\partial}{\partial x} + \sin\theta\frac{\partial}{\partial y}$$

$$\hat{\theta} = -\sin\theta\frac{\partial}{\partial x} + \cos\theta\frac{\partial}{\partial y}$$

Comparing with the basis from (a):

$$\frac{\partial}{\partial r} = \frac{\partial x^i}{\partial r}\frac{\partial}{\partial x^i} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial}{\partial y} \quad \| \ \frac{\partial x}{\partial r} = \cos\theta$$

$$\frac{\partial}{\partial r} = \cos\theta\frac{\partial}{\partial x} + \sin\theta\frac{\partial}{\partial y} \quad \Rightarrow \quad \hat{r} = \partial/\partial r$$

$$\frac{\partial}{\partial \theta} = \left(\frac{\partial x}{\partial \theta}\right)\frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta}\frac{\partial}{\partial y} \quad \| \ \frac{\partial x}{\partial \theta} = -r\sin\theta$$

$$= -r\sin\theta\frac{\partial}{\partial x} + r\cos\theta\frac{\partial}{\partial y} \quad \Rightarrow \quad \hat{\theta} = \frac{1}{r}\frac{\partial}{\partial \theta}$$

The coordinates $\hat{r}$, $\hat{\theta}$ in terms of $\partial/\partial r$ & $\partial/\partial\theta$ are

$$\begin{cases} \hat{r} = \partial/\partial r \\ \hat{\theta} = \frac{1}{r}\partial/\partial\theta \end{cases}$$

Now let's get g in $(\hat{r}, \hat{\theta})$ basis.

$$g_{\hat{r}\hat{r}} = g(\hat{r}, \hat{r}) = g(\cos\theta\,\hat{x} + \sin\theta\,\hat{y}, \ \cos\theta\,\hat{x} + \sin\theta\,\hat{y})$$

$$= \cos^2\theta\,\underbrace{g(\hat{x}, \hat{x})}_{=1} + 2\sin\theta\cos\theta\,\underbrace{g(\hat{x}, \hat{y})}_{=0} + \sin^2\theta\,\underbrace{g(\hat{y}, \hat{y})}_{=1}$$

$$g_{\hat{r}\hat{r}} = 1$$

here we used the fact that g is bilinear function of two vectors. Similarly

$$g_{\hat{r}\hat{r}} = g_{\hat{r}\hat{r}} = g(\cos\theta\,\hat{x} + \sin\theta\,\hat{y},\ -\sin\theta\,\hat{x} + \cos\theta\,\hat{y})$$

$$= -\cos\theta\sin\theta\ \underbrace{g(\hat{x},\hat{x})}_{=1} + \cos\theta\sin\theta\ \underbrace{g(\hat{y},\hat{y})}_{=1} + \text{terms} \cdot \underbrace{g(\hat{x},\hat{y})}_{=0}$$

$$= 0$$

$$g_{\hat{\theta}\hat{\theta}} = g(-\sin\theta\,\hat{x} + \cos\theta\,\hat{y},\ -\sin\theta\,\hat{x} + \cos\theta\,\hat{y})$$

$$= \sin\theta\ \underbrace{g(\hat{x},\hat{x})}_{=1} + \cos\theta\ \underbrace{g(\hat{y},\hat{y})}_{=1} + \text{terms} \cdot \underbrace{g(\hat{x},\hat{y})}_{=0}$$

$$= 1$$

$\Rightarrow$ in the $\{\hat{r},\hat{\theta}\}$ basis the metric g is

$$g = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Exercise 2.16

df in the cardinal basis $(\partial/\partial x, \partial/\partial y)$ has the form

$$df = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)$$

One forms transform by $\Lambda^{i}_{i'}$ the inverse matrix of the one for vector transformation: $\Lambda^{i}_{i'} = \frac{\partial x^i}{\partial x^{i'}}$ where primed coords are the new coords

$\Rightarrow$ $df_i = \frac{\partial f}{\partial x^i}$

$$\Rightarrow \tilde{df}_{i'} = \Lambda^{i}_{i'}\,\frac{\partial f}{\partial x^i} = \frac{\partial x^i}{\partial x^{i'}}\,\frac{\partial f}{\partial x^i}$$

$$\Rightarrow \tilde{df}_r = \frac{\partial x}{\partial r}\frac{\partial f}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial f}{\partial y} = 2\,\frac{\partial f}{\partial x} \quad \text{by chain rule}$$

$$\tilde{df}_\theta = \frac{\partial x}{\partial \theta}\frac{\partial f}{\partial x} + \frac{\partial y}{\partial \theta}\frac{\partial f}{\partial y} = 2\,\frac{\partial f}{\partial \theta}$$

$$\Rightarrow \tilde{df} = \left(\frac{\partial f}{\partial r}, \frac{\partial f}{\partial \theta}\right)$$ 4- this makes sense since we in coordinate basis

As for $\tilde{df}$ we can just use g in polar coords

$g = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \Rightarrow \tilde{df} = g^{ij}\, \tilde{df}_i$ // $(g^{ij})^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & r^{-2} \end{pmatrix}$

$\tilde{df} = (\tilde{df}_1, \frac{1}{r^2}\, \tilde{df}_2) = (\frac{\partial f}{\partial r}, r^{-2}\, \frac{\partial f}{\partial \theta})$

Now lets go to non-coordinate basis $(\hat{r}, \hat{\theta})$

// $\hat{r} = \cos\theta\, \hat{x} + \sin\theta\, \hat{y}$, $\hat{\theta} = -\sin\theta\, \hat{x} + \cos\theta\, \hat{y}$

$\hat{r} = \partial/\partial r$, & $\hat{\theta} = \frac{1}{r}\partial/\partial\theta$

$\Rightarrow \tilde{df}(\hat{x}) = \frac{\partial f}{\partial x}$, $\tilde{df}(\hat{y}) = \frac{\partial f}{\partial y}$

in same vein: $\tilde{df}(\frac{\partial}{\partial r}) = \frac{\partial f}{\partial r}$, $\tilde{df}(\frac{\partial}{\partial \theta}) = \frac{\partial f}{\partial \theta}$

$\Rightarrow \tilde{df}(\hat{r}) = \tilde{df}(\cos\theta\, \hat{x} + \sin\theta\, \hat{y}) = \cos\theta\, \tilde{df}(\hat{x}) + \sin\theta\, \tilde{df}(\hat{y})$

$= \cos\theta\, \frac{\partial f}{\partial x} + \sin\theta\, \frac{\partial f}{\partial y} = (\cos\theta\, \frac{\partial}{\partial x} + \sin\theta\, \frac{\partial}{\partial y}) f$

= But we can also write this in polar coords (r, θ):

$\tilde{df}(\hat{r}) = \tilde{df}(\partial/\partial r) = \frac{\partial f}{\partial r}$ (same same which is expected)

$\tilde{df}(\hat{\theta}) = \tilde{df}(\frac{1}{r}\frac{\partial}{\partial \theta}) = \frac{1}{r}\frac{\partial f}{\partial \theta}$ (rescaled by $\frac{1}{r}$, expected)

$\Rightarrow \tilde{df} = (\frac{\partial f}{\partial r}, \frac{1}{r}\frac{\partial f}{\partial \theta})$ in $\{\hat{r}, \hat{\theta}\}$ basis

$\Rightarrow \tilde{df} = g^{ij}\, \tilde{df}_i = (\frac{\partial f}{\partial r}, \frac{1}{r}\frac{\partial f}{\partial \theta})$ // g^{ij} then is for g^{-1} in $(\hat{r}, \hat{\theta})$ basis which is $\mathbb{I}$